Fusing Rankings: A Practical Look at Hybrid Retrieval for Grounded Assistants

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Demo document written for PaperSensei. The experiment is illustrative and not a published result.

Abstract

Assistants that answer from documents fail in two familiar ways: they miss the passage that holds the answer, or they find it and then say something the passage does not support. This paper looks at the first failure. We compare keyword search, vector search and a simple fusion of the two called reciprocal rank fusion. On a small illustrative benchmark, fusion finds the right passage more often than either method alone, needs no training and adds almost no latency. We explain why it works, when it does not, and how to tell the difference in production.

1 Introduction

Retrieval is the quiet half of a grounded assistant. A language model can only cite what it is shown, so if the right passage never reaches the prompt, no amount of careful prompting will produce a faithful answer.

Two families of retrieval dominate practice. Keyword search scores a passage by how many of the query's words it contains, weighted by how rare those words are. Vector search embeds the query and every passage as points in a shared space and returns the nearest neighbours. Each is strong where the other is weak, which suggests combining them.

Our contribution is practical rather than novel: a clear account of reciprocal rank fusion, a small experiment that shows its effect, and a checklist for deciding whether hybrid retrieval is worth it for a given corpus.

2 Background

BM25 is the standard keyword scoring function. It rewards a passage for containing query terms, rewards rare terms more than common ones through inverse document frequency, and saturates so that repeating a word many times stops helping. It is exact: a query for an error code finds that code.

Dense retrieval uses a neural encoder to map text to vectors so that passages with similar meaning sit close together, even when they share no words. A question about cancelling a subscription can match a passage about ending a plan. Its weakness is the opposite of BM25: rare identifiers, names and numbers are often blurred together.

3 Method

Reciprocal rank fusion merges ranked lists without looking at their scores. Each passage receives, from every list it appears in, a contribution of one divided by the sum of a constant k and its rank in that list. The contributions are added and the passages are sorted by the total.

Ignoring raw scores is the point. BM25 scores and cosine similarities live on different scales, and normalising them is fragile. Ranks are always comparable. The constant k, usually 60, flattens the advantage of the very top positions, so a passage ranked well by both methods beats one ranked first by

only one of them.

In our setup each method returns its top 20 passages, fusion keeps the best 6, and those 6 are given to the model as numbered sources it must cite.

4 Experiments

We built an illustrative benchmark of 120 questions over a corpus of 900 passages drawn from product documentation, release notes and support articles. Each question has one passage marked as the answer. We report recall at 6: the share of questions whose answer passage is among the 6 passages given to the model.

We compare three systems: BM25 alone, a small open embedding model alone, and reciprocal rank fusion of the two with k set to 60. All three run on a laptop CPU.

5 Results

On this benchmark BM25 alone reached a recall at 6 of 0.71 and vector search alone reached 0.78. Fusion reached 0.89. Fusion added under 5 milliseconds per query, almost all of it the second search.

The gains were not uniform. Questions containing exact identifiers such as version numbers were answered best by BM25, and vector search often missed them. Paraphrased questions were answered best by vector search. Fusion recovered most of both groups, which is why its total is higher than either.

6 Limitations

The benchmark is small and illustrative, so the numbers show the shape of the effect rather than its size. Fusion also cannot rescue a passage that neither method retrieves; if both lists miss it, the answer is still lost. Finally, fusion treats both methods as equally trustworthy. When one is much weaker on a corpus, weighting the lists or dropping the weak method can do better.

7 Conclusion

Hybrid retrieval with reciprocal rank fusion is cheap, robust and easy to reason about. For assistants that must cite their sources, it is a sensible default: measure recall on your own questions, keep both rankings visible for debugging, and only reach for learned rerankers once fusion has stopped improving.

References

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